

When Front-Stage Innovation Overloads Back-Stage Workers: A Dynamic Model of Workforce, Capability, and Customer Trust in Omnichannel Retail

(Authors' names are not included for peer review)

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Abstract. Problem definition. Digital investment in omnichannel retail generates back-stage demand on workers that standard dashboards cannot see. Front-stage automation amplifies operational complexity faster than back-stage capacity can absorb it, creating a structural measurement gap: technology returns and labour consequences appear on different ledgers, and the workforce that determines whether technology delivers its designed capability goes systematically underinvested.

Methodology/results. We formalise this gap as a parsimonious dynamic model and derive three analytically characterised results. (i) A computable digitalisation ceiling exists beyond which additional front-stage investment destroys customer lifetime value (CLV) by overloading back-stage workers; revenue metrics continue rising even as CLV collapses. (ii) The entire return to workforce investment flows through service capacity, invisible to dashboards that treat payroll as a cost line. (iii) The system is bistable: the neglected firm faces a safe rollout window less than two-fifths the size of its investing peer. We provide suggestive empirical support using panel data on 13 US and 11 European retailers (2010–2024): the technology–workforce interaction is positive in cross-section, and an inverted-U between e-commerce share and customer satisfaction aligns with the predicted digitalisation ceiling.

Managerial implications. The model supplies two computable numbers absent from standard dashboards: a digitalisation ceiling and a maximum safe rollout size. Both are workforce-contingent, making workforce investment a structural requirement of digital strategy rather than a social concession.

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1. Introduction

Omnichannel grocery retailers have invested heavily in digital capabilities in recent years (Hübner et al. 2021, Bijmolt et al. 2021), enabling customers to build baskets across channels, reserve items for same-day collection, and track substitutions in real time. Yet as digital engagement has grown, retailers have encountered a structural asymmetry in how costs and returns are measured. Technology investment registers as successful in the front-stage domain where it is evaluated, while the back-stage workforce that must fulfil the digitally generated demand absorbs consequences that appear nowhere in the technology business case (Corbett 2024).

The asymmetry is structural, not incidental. Front-stage digital capabilities generate customer journeys that multiply handoff complexity far faster than order volume grows (Saghiri et al. 2017), placing distinctive operational demands on back-stage workers (pickers, drivers, click-and-collect associates). The empirical literature confirms the scale of these consequences: workforce investment generates returns that are both large and invisible to standard metrics (Fisher et al. 2021, Kesavan et al. 2022), and firms that invest in workforce capacity outperform on both operational and financial measures (Ton 2014), yet the causal chain runs through mechanisms that standard dashboards do not monitor.

What is missing is a formal account of the mechanism. The empirical literature establishes that workforce investment pays off; it does not characterise *when* additional front-stage investment begins to destroy rather than create value, *how fast* a firm can scale digitalisation before the back-stage workforce is overwhelmed, or *what dynamic relationship* connects technology investment, workforce well-being, and customer trust over time. These are not questions that regression analysis on historical data can answer: they require a model that treats workforce well-being and customer trust as co-evolving state variables responding to a controllable investment sequence (see §2).

This paper formalises the economic structure of this problem as a parsimonious dynamic model. Technology and workforce are not independent inputs to service capability but strategic complements (Milgrom and Roberts 1990, Bainbridge 1983, Kremer 1993). The central modelling innovation is the *workforce capacity multiplier*: effective service capability is the product of the technology stock and a workforce engagement index. When front-stage demand grows faster than back-stage workforce capacity, a service gap opens that depletes workforce well-being, which reduces effective capability, which widens the gap further: a reinforcing deterioration that single-domain models cannot generate.

Three formally characterised results follow. A *Front-Stage Trap* (Theorem 1): a computable digitalisation threshold exists beyond which additional front-stage investment reduces customer lifetime value by overloading back-stage workers. A *Good Jobs Buffer* (Theorem 2): the return to workforce investment flows through a capacity-level channel that is invisible to dashboards measuring only operational sensing. A *Burnout Cliff* (Theorem 3): the system exhibits bistability, and a firm that neglects its workforce faces a safe deployment window less than two-fifths the size of one that invests in its people. A fourth result, Proposition 1, establishes that the marginal value of workforce investment is increasing in digitalisation level. Workforce investment is therefore not a social concession but a structural requirement of digital strategy.

2. Related Literature

This paper sits at the intersection of three research streams. We review each and identify the gaps the model addresses.

Service quality erosion. The foundational insight that service quality can erode endogenously under demand pressure originates with Oliva and Serman (2001), whose system-dynamics model (Serman 2000) produces death spirals in which workers cut corners under demand pressure. Their finding that workers are roughly twice as willing to cut corners as to work overtime grounds the overload-asymmetry parameter σ in the present model. Subsequent work extends the logic to capability erosion (Rahmandad and Repenning 2016, Repenning and Serman 2001), e-commerce growth (Oliva et al. 2003), and regulatory compliance (Martinez-Moyano et al. 2014); KC and Terwiesch (2009) confirm that sustained high workload degrades service outcomes empirically, and Corbett (2024) calls for formal dynamic models of workforce well-being in operations. **Gap 1:** no model in this tradition treats workforce well-being as a positive capability multiplier. **Gap 2:** no Oliva–Serman extension adds customer trust, a front-/back-stage split, or digital scaling.

Workforce investment and retail performance. A substantial empirical literature documents the link between workforce practices and performance. The Good Jobs Strategy (Ton 2014) identifies four operational choices that drive outperformance; causal evidence comes from field experiments showing that responsible scheduling raises productivity and sales (Kesavan et al. 2022) and that profit-maximising staffing exceeds prevailing practice (Fisher et al. 2021, Dubé et al. 2020). The conceptual mechanism is the *service profit chain* (Heskett et al. 1994), which has not been formalised as a dynamic model with computable boundary conditions. The Job Demands-Resources model (Bakker and Demerouti 2007, Demerouti et al. 2001) provides the micro-foundation: overload depletes well-being faster than equivalent surplus restores it, and productivity responses

Table 1 Literature gaps and model contributions.

Gap	Closest prior work	This paper
1. No model treats W as a positive capability multiplier	Oliva and Sterman (2001): W as degradation only	Effective capability as the product of technology and workforce engagement
2. No Oliva–Sterman extension adds trust, domain split, or digital scaling	Oliva and Sterman (2001); Martinez-Moyano et al. (2014)	Customer trust as a co-evolving state; front/back domain split; digital scaling
3. Good Jobs Strategy / service profit chain (SPC) unformalised	Ton (2014); Heskett et al. (1994): empirical only	Dynamic formalisation with analytical equilibria and boundary conditions
4. No cross-domain front-to-back overload model	Saghiri et al. (2017): conceptual only	Super-linear back-stage demand as a function of digital depth and customer trust
5. Digitalisation-workforce cross-partial not derived for services	No direct service prior; see also Gans and Goldfarb (2026) (O-ring)	Formally derived: the return to workforce investment rises with the technology base

scale with engagement (Mas and Moretti 2009). These two features (asymmetric overload and engagement-scaled sensitivity) underpin the workforce dynamics in §3.3. **Gap 3:** no dynamic model formalises the Good Jobs Strategy or the service profit chain with computable boundary conditions.

Omnichannel capability. Digitalisation creates considerable challenges for retail logistics, with fulfilment complexity scaling super-linearly as channel combinations increase (Hübner et al. 2021, Saghiri et al. 2017, Bell et al. 2018, Acimovic and Graves 2015). Automation shifts the operator's role toward exception handling (Bainbridge 1983), making the quality of residual human tasks more consequential for overall output (Gans and Goldfarb 2026). Technology–workforce strategic complementarity (Milgrom and Roberts 1990, Aral et al. 2012) motivates the model's central structural assumption that the marginal value of workforce engagement rises with the technology base (§3.1). **Gap 4:** no model formalises front-to-back cross-domain overload. **Gap 5:** the digitalisation-workforce cross-partial has not been derived for services.

Table 1 summarises the five gaps and the model's response to each.

3. Model

Sections 3.1–3.5 formalise the workforce capacity multiplier introduced in §1. The front-stage digitalisation level $F > 0$ is treated as an exogenous managerial control: it sets the complexity and volume of customer-facing digital features, and thereby determines the demand placed on the back-stage operation. The model traces the dynamic consequences of choosing F for the three co-evolving back-stage state variables and, through them, for customer lifetime value (CLV). Each of

the eight free parameters has a single, distinct role with a clean kill condition (Table 2): setting it to its neutral value eliminates exactly one mechanism, allowing the model to nest single-domain benchmarks as special cases.

3.1. Stocks and Effective Capability

Three state variables evolve jointly: $K(t) > 0$, the back-stage technology stock (installed infrastructure, warehouse management systems, and process standardisation); $W(t) \in (0, 1)$, an index of workforce well-being (engagement, energy, and cognitive capacity among pickers, drivers, and click-and-collect associates); and $T(t) \in (0, 1)$, customer trust (the customer base's aggregate confidence that the retailer will deliver reliably). All three are dimensionless and normalised to their natural operating ranges. Throughout, starred variables (K^* , W^* , T^*) denote steady-state equilibrium values; the pre-shock (baseline) digitalisation level is written F_0 ; and ΔF denotes an upgrade step, the increase in F from F_0 .

The central structural innovation is the *effective capability*. A warehouse management system or substitution-handling process can only deliver its designed performance if the workforce operating it has the capacity to do so. Effective capability is therefore not the technology stock alone but its product with a workforce engagement multiplier:

$$K_{\text{eff}}(t) = K(t) \cdot h(W(t)),$$

$$h(W) = (1 - p) + pW, \quad p \in (0, 1). \quad (1)$$

Effective capability is the product of the technology stock and a workforce capacity multiplier $h(W)$. The multiplier ranges from $h(0) = 1 - p$ (at zero well-being, the workforce delivers only the irreducible, automation-driven fraction $1 - p$ of the technology's designed potential) to $h(1) = 1$ (a fully engaged workforce realises the technology's full design capacity). The cross-partial

$$\frac{\partial^2 K_{\text{eff}}}{\partial K \partial W} = p > 0 \quad (2)$$

is strictly positive for all $p > 0$, establishing technology and workforce as complements in the Milgrom and Roberts (1990) sense.

The parameter p governs complementarity: $p = 0$ decouples workforce from capability; $p = 1$ makes the operation fully labour-intensive. The baseline $p = 0.5$, motivated by capital-skill complementarity evidence (Krusell et al. 2000), represents a moderate starting point; estimation is discussed in §7.2. All three theorems hold qualitatively for any $p \in (0, 1)$.

3.2. Demand and the Service Gap

Customer demand on back-stage operations is

$$D(T, F) = (1 + aT) \cdot F^b, \quad (3)$$

where $a > 0$ is the trust-demand coupling and $b > 1$ is the front-stage amplification exponent. The trust-demand coupling reflects the established finding that customer trust drives spending consolidation and order frequency (Sirdeshmukh et al. 2002, McKnight et al. 2002). The super-linearity $b > 1$ captures the well-documented increase in back-stage fulfilment complexity as front-stage channel breadth grows: as shown in §2, Bell et al. (2018) and Acimovic and Graves (2015) document patterns consistent with fulfilment error rates and demand-capacity mismatches rising faster than linearly with channel combinations. We set $b = 1.3$ as an illustrative, structurally conservative baseline motivated by this literature; the Front-Stage Trap requires only $b > 1$, and all three theorems hold qualitatively across $b \in [1.1, 2.0]$ (EC Table EC.3).

The *service gap* measures the shortfall between demand and effective capability:

$$G(t) = D(T(t), F) - K(t) \cdot h(W(t)). \quad (4)$$

When $G > 0$ (overload), customers experience failures. When $G < 0$ (surplus), service runs reliably. We write $[G]^+ = \max(G, 0)$ and $[-G]^+ = \max(-G, 0)$ for the positive and negative parts.

3.3. Dynamics

The feedback loops in §3.5 are built from the three ordinary differential equations (ODEs) introduced below, which translate each economic mechanism into a rate equation. All rate parameters carry units of months⁻¹; each ODE is piecewise-smooth, activating different terms in the overload regime ($G > 0$, so $[G]^+ = G$, $[-G]^+ = 0$) and the surplus regime ($G < 0$, so $[G]^+ = 0$, $[-G]^+ = -G$).

Capability. The technology stock evolves as

$$\frac{dK}{dt} = r [G]^+ W - \delta K + u, \quad (5)$$

where $r > 0$ is the sensing-adaptation rate, $\delta > 0$ is the depreciation rate, and $u \geq 0$ is direct back-stage investment. The sensing term $r [G]^+ W$ captures organisational learning from service failures (Tucker 2007): the product structure requires both a gap to detect ($[G]^+ > 0$) and workforce capacity to act on it ($W > 0$), reflecting the finding that capability accumulation scales with the workforce knowledge stock (Argote and Epple 1990).

Workforce well-being. Well-being rises under surplus capacity and direct investment; it falls through natural decay and overload-amplified depletion:

$$\frac{dW}{dt} = ([-G]^+ + f)(1 - W) - (\mu + \sigma [G]^+)W. \quad (6)$$

Two grouped terms ensure $W(t) \in [0, 1]$. The first drives W upward (vanishing at $W = 1$): capability surplus and direct workforce investment raise well-being with diminishing returns, consistent with the JD-R model (Bakker and Demerouti 2007). The second drives W downward (vanishing at $W = 0$): natural decay (μ) and overload erosion ($\sigma [G]^+$) deplete well-being, with $\sigma > 1$ reflecting the asymmetry documented by Oliva and Sterman (2001). All three theorems hold for any $\sigma > 1$ (EC Table EC.3).

Customer trust. Trust builds through reliable service and erodes faster when reliability fails (Sirdeshmukh et al. 2002, Mittal et al. 1998):

$$\frac{dT}{dt} = \beta [-G]^+ (1 - T) - \eta \beta [G]^+ T - \lambda T, \quad (7)$$

where $\beta > 0$ is the trust-update rate, $\eta > 1$ is the erosion asymmetry, and $\lambda > 0$ is natural trust decay. We set $\eta = 2.0$, consistent with the asymmetric coefficients reported by Sirdeshmukh et al. (2002) (approximately 2.5:1 depletion-to-enhancement ratio).

The system (5)–(7) is piecewise-smooth; existence and uniqueness of solutions follow from Filippov's (1988) framework (details and smoothing robustness in EC §EC.1.1 and EC.1.8).

3.4. State-Space Invariance and Empirical Mapping

The state space $K > 0$, $W \in [0, 1]$, $T \in [0, 1]$ is forward-invariant for all $t \geq 0$; the verification via directional derivatives at each boundary is given in EC §EC.1.1.

Observable field proxies include: app feature index for front-stage capability (F), employee engagement composite for workforce well-being (W), net promoter score for customer trust (T), and on-time fulfilment shortfall for the service gap (G). Parameter calibration details are in EC §EC.4.

3.5. Feedback-Loop Architecture

Figure 1 shows the causal-loop architecture of the three ODEs, using the conventions of Sterman (2000). The diagram contains four feedback loops, each traceable from the state-variable arrows.

Loop R1 ($T \rightarrow D \rightarrow G \rightarrow T$, reinforcing): trust raises demand, which changes the service gap, feeding back to trust. **Loop R2** ($G \rightarrow W \rightarrow K_{\text{eff}} \rightarrow G$, reinforcing): overload depletes well-being, reducing effective capability and widening the gap further — the deterioration loop behind the

Figure 1 Causal-loop architecture. Solid boxes are state variables; dashed boxes are exogenous controls. Arrow signs indicate direction of effect; loop polarity is shown in parentheses. R2 and R3, created by the capacity multiplier $h(W)$, are novel and do not exist in single-domain models.

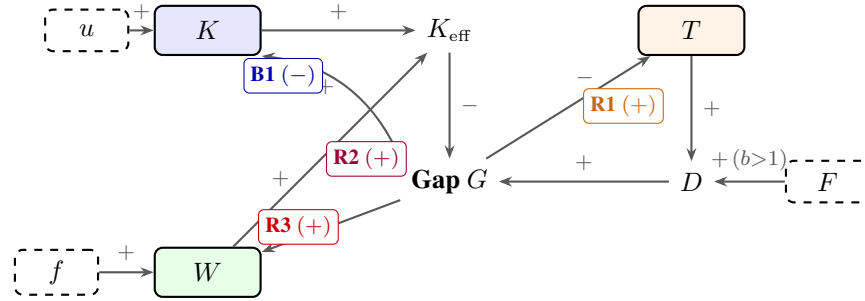


Table 2 Model parameters, baseline values, and kill conditions.

#	Parameter	Baseline	Timescale	Kill	Mechanism killed	Empirical anchor
1	b (demand super-linearity)	1.3	–	$b=1$	Front-Stage Trap	Bell et al. (2018); Acimovic & Graves (2015)
2	a (trust-demand coupling)	0.4	–	$a=0$	Demand feedback	McKnight et al. (2002); Sirdeshmukh et al. (2002)
3	p (workforce complementarity)	0.5	–	$p=0$	Capacity multiplier	Krusell et al. (2000)
4	σ (overload asymmetry)	2.0	–	$\sigma=1$	Tipping point	Oliva and Sterman (2001); Bakker and Demerouti (2007) [†]
5	η (erosion asymmetry)	2.0	–	$\eta=1$	Trust asymmetry	Sirdeshmukh et al. (2002)
6	r (sensing-adaptation)	0.4	$\tau_r \approx 1.7$ mo	$r=0$	Self-correction dies	Tucker (2007); Argote & Epple (1990)
7	δ (depreciation)	0.03	$\tau_\delta \approx 23$ mo	–	–	Illustrative (EC §EC.4)
8	β (trust-update rate)	0.12	$\tau_\beta \approx 5.8$ mo	$\beta \rightarrow 0$	Trust frozen	Illustrative (EC §EC.4)
+	μ (natural W decay)	0.05	$\tau_\mu \approx 14$ mo	$\mu=0$	W exponential only	Demerouti et al. (2001); Dubé et al. (2020)
+	λ (natural T decay)	0.025	$\tau_\lambda \approx 28$ mo	$\lambda=0$	T exponential only	Illustrative (EC §EC.4)

All time units are months. Timescales are $\ln(2)/\theta$ (half-life). Eight free parameters (rows 1–8); μ and λ govern natural decay. “Illustrative” parameters are calibrated within empirically informed ranges (EC §EC.4). [†] Magnitude comparator from professional services; see §3.3. Direct estimation of b and p from retailer panel data is the primary empirical extension (§7.2).

Front-Stage Trap. Loop R3 ($W \rightarrow K_{\text{eff}} \rightarrow G \rightarrow W$, reinforcing): capability degradation escalates workforce depletion — the loop behind the Burnout Cliff. **Loop B1** ($G \rightarrow K \rightarrow K_{\text{eff}} \rightarrow G$, balancing): overload triggers sensing investment that raises capability and closes the gap. Loops R2 and R3, created by the capacity multiplier $h(W)$, have no counterpart in prior quality-erosion models.

3.6. Parameters and Special Cases

Table 2 summarises all parameters. Two control variables complete the model: $u \geq 0$ (capability investment) and $f \geq 0$ (workforce investment). The model recovers the core dynamics of Oliva and Sterman (2001) when $p = 0$ and $a = 0$ (see EC §EC.2 for the structural mapping), and reduces to a Solow-type capital-accumulation equation when $p = 0$, $a = 0$, and $r = 0$.

4. Analytical Results

We derive four formal results: Theorems 1–3 characterise three phenomena that no single-domain model can produce; Proposition 1 establishes digitalisation-workforce complementarity. Full proofs are in the Electronic Companion (§EC.1); here we provide proof sketches.

4.1. Equilibrium Structure

The model generates two distinct long-run outcomes. The *surplus equilibrium*: effective capability exceeds demand, the workforce is healthy, trust builds, and the firm settles into a self-reinforcing high-performance state. The *low-trust attractor*: demand has overwhelmed capability, the workforce is depleted, and customers have withdrawn trust; exit requires structural intervention.

The technical characterisation of these two states is as follows. The surplus equilibrium exists for all $F < F_{\text{crit}}$ and is unique under Condition C2 (EC Lemma 2, §EC.1.2). When effective capability exceeds demand, define the operational surplus $S^* \equiv K^*h(W^*) - D(T^*, F) \geq 0$ as the excess of effective capability over demand at equilibrium. Equilibrium trust is an increasing function of S^* , saturating toward one as surplus grows: $T^* = \beta S^* / (\beta S^* + \lambda)$. Equilibrium well-being is similarly increasing in surplus and investment: $W^* = (S^* + f) / (S^* + f + \mu)$. The capability stock settles at $K^* = u/\delta$, determined entirely by the investment-depreciation balance. Full algebraic derivations are in EC §EC.1.2. Both T^* and W^* are increasing in S^* , so any improvement in effective capability propagates into higher trust and well-being at the equilibrium, creating the reinforcing dynamics that underpin the Good Jobs Buffer.

The low-trust attractor is reached whenever demand persistently overwhelms capability. With $T^* \approx 0$, demand collapses to its digitalisation-driven floor and the workforce settles at a low but strictly positive well-being level, held above zero by the workforce investment rate f . Effective capability at this attractor falls short of demand, maintaining a persistent residual gap. The attractor exists whenever $F^b > (u/\delta) h(W_{\text{low}}^*)$ and is locally stable (EC Remark 1).

Maintained conditions. The formal results hold under four regularity conditions (C1)–(C4), verified analytically or numerically across the parameter space (EC Remark 3): surplus equilibrium existence, uniqueness, regularity, and low-trust attractor existence (EC §EC.1.2).

REMARK 1 (ANALYTICAL TRANSPARENCY). The table below classifies each result by evidentiary basis: Tier A (analytic for all valid parameters), Tier B (analytic under C1–C4), or Tier C (computational verification).

Result	Tier	Basis
Thm 1(i): equilibrium existence/nonexistence	A	Analytic (IVT + $\Phi(0)$ sign)
Thm 1(ii): trajectories bounded	A	Analytic (Gronwall; Lemma 3)
Thm 1(ii): wide-basin convergence	C	Numerical grid, $K_0 = K^*$ slice
Lemma 2 (uniqueness of S^*)	B	Analytic under C2-SC ($\beta(f+\mu) < \lambda$; holds throughout parameter space; EC §EC.1.2)
Thm 2: $\partial \text{CLV}^*/\partial f > 0$ (sign)	B	IFT chain; GE feedback correction sign analytic
Thm 2: GE feedback magnitude (≈ 0.09)	C	Confirmed across 100 configurations
Thm 3(i): $\Delta F_{\text{collapse}}$ threshold	A	Direct from Thm 1
Thm 3(i): convergence from pre-shock IC	C	Numerical; basin-dependent
Prop 1 cross-partial	A	Analytic ($K_{\text{eff}} = K h(W)$)
Prop 1 $F_{\text{max}}(W)$ boundary formula	A	Exact at $S^* \rightarrow 0$
Prop 1 $F_{\text{max}}(W)$ interior upper bound	B	Analytic sign; $\approx 8\%$ correction numeric
Corollary 5, Case 1	B	Thm 1 + Thm 2 directly
Corollary 5, Case 2 near B_{min}	B	Continuity from Case 1 (IFT; EC §EC.1.7)
Corollary 5, Case 2 interior	C	Computational; $c_F/c_f \in [0.5, 5]$

4.2. Theorem 1: The Front-Stage Trap

THEOREM 1 (Front-Stage Trap). *Suppose $b > 1$ and $p > 0$. Define the digitalisation threshold*

$$F_{\text{crit}} = \left(\frac{u}{\delta} \cdot \left[(1-p) + p \frac{f}{f+\mu} \right] \right)^{1/b}, \quad (8)$$

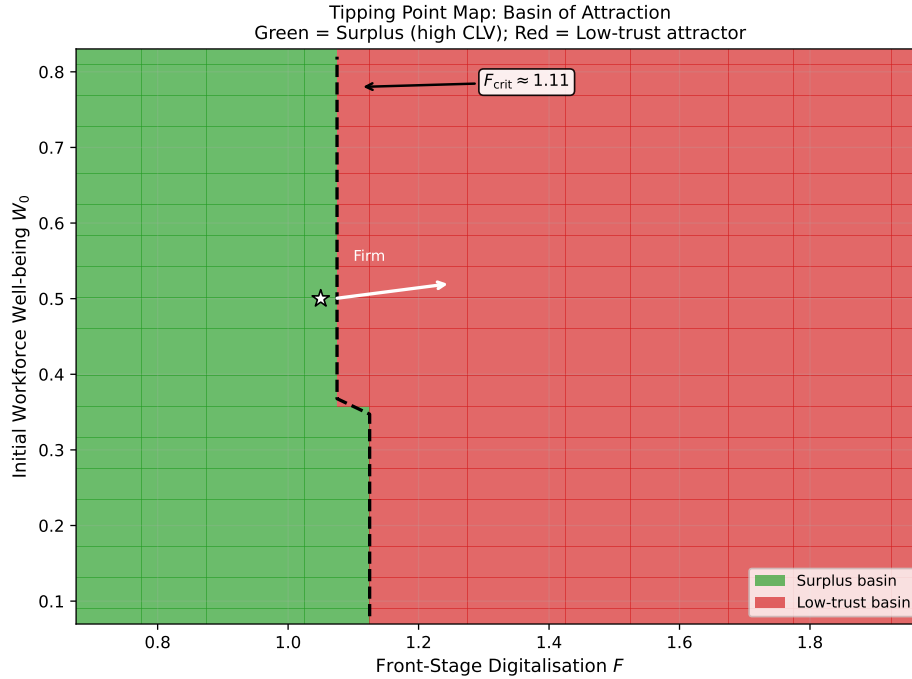
which depends only on exogenous parameters $(u, \delta, p, f, \mu, b)$ and equals $K^ h(f/(f+\mu))^{1/b}$, the capacity multiplier evaluated at the boundary workforce level. Then:*

(i) For $F < F_{\text{crit}}$, the surplus equilibrium (unique under C2; Lemma 2, EC §EC.1.2) exists with $T^ > 0$ and $\text{CLV}^* > \text{CLV}_{\text{low}}$ (the CLV at the low-trust attractor).*

(ii) For $F \geq F_{\text{crit}}$, no surplus equilibrium exists. All forward trajectories remain in a bounded compact set (Lemma 3, EC §EC.1.2), and the low-trust attractor is locally stable (EC Remark 1). The threshold F_{crit} is decreasing in p and increasing in f : the Trap binds earlier when workforce-technology coupling is strong, and is relaxed by workforce investment.

Proof sketch. The surplus equilibrium requires $\Phi(0) > 0$ (where $\Phi(S) \equiv K^* h(W^*(S)) - D(T^*(S), F) - S$ is the surplus residual; full construction in EC §EC.1.2), which holds if and only if $F < F_{\text{crit}}$. At $F = F_{\text{crit}}$, the surplus $S^* = 0$ and the equilibrium ceases to exist. For $F > F_{\text{crit}}$, $G > 0$, activating overload depletion of W through Eq. (6), which reduces $h(W)$, which further widens G through Loop R2. The reinforcing dynamics drive the system toward the low-trust attractor. That trajectories remain bounded for all t follows from Lemma 3 (EC §EC.1.2): under overload, $\dot{K} \leq C(F) - \delta K$ where $C(F) = r(1+a)F^b + u$ is a demand-capacity bound, giving $K(t) \leq \max(K_0, C(F)/\delta)$ by Gronwall's lemma; together with $W, T \in [0, 1]$ by construction, the state space is forward-invariant and compact. Full proof in EC §EC.1.3. $\square$

Figure 2 Basin-of-attraction map. Green cells converge to the surplus attractor (high CLV) for $F < F_{\text{crit}} \approx 1.11$; red cells converge to the low-trust attractor for $F \geq F_{\text{crit}}$. Doubled-grid validation confirms boundary stability (EC §EC.4.2).



At the calibrated baseline, $F_{\text{crit}} \approx 1.11$, corresponding to roughly a 40% increase in digital feature complexity above $F_0 = 0.80$. Figure 2 maps the regime structure: green cells converge to the surplus attractor (high CLV); red cells converge to the low-trust attractor (CLV ≈ 400). The cross-domain CLV sign reversal requires $b > 1$ (super-linear demand) and $p > 0$ (workforce-contingent threshold); at $p = 0$ the threshold reduces to $(u/\delta)^{1/b}$ and no longer responds to workforce investment.

4.3. Theorem 2: The Good Jobs Buffer

THEOREM 2 (Good Jobs Buffer). *Suppose $p > 0$ and the surplus equilibrium exists and is stable (Conditions C1–C4, i.e., the surplus equilibrium exists, is unique, and the low-trust attractor is present; Remark 1). Workforce investment f raises steady-state CLV through a multiplicative capacity-level channel:*

$$\frac{\partial \text{CLV}^*}{\partial f} > 0 \quad \text{for all } f \geq 0. \quad (9)$$

The return decomposes into a direct capacity effect and an indirect trust effect:

$$\begin{aligned} \frac{\partial \text{CLV}^*}{\partial f} = & \underbrace{\frac{\partial \text{CLV}}{\partial T^*} \cdot \frac{\partial T^*}{\partial S^*} \cdot K^* \cdot p}_{\text{direct: } h(W) \text{ raises } K_{\text{eff}}} \cdot \frac{\partial W^*}{\partial f} \\ & + \underbrace{\text{higher-order terms via sensing}}_{\text{indirect: } W \text{ in } r[G]^+ W}. \end{aligned}$$

Proof sketch. Since $W^* = (S^* + f)/(S^* + f + \mu)$ at the surplus equilibrium, differentiating gives $\partial W^*/\partial f = \mu/(S^* + f + \mu)^2 > 0$: workforce investment raises steady-state well-being. Higher W^* raises $h(W^*) = (1 - p) + pW^*$, which increases $K_{\text{eff}} = K^* \cdot h(W^*)$, which widens the surplus S^* , which raises $T^* = \beta S^*/(\beta S^* + \lambda)$, which raises CLV. Each link is strictly positive: $\partial h/\partial W = p > 0$, $\partial S/\partial h = K^* > 0$, $\partial T^*/\partial S^* > 0$, $\partial \text{CLV}/\partial T^* > 0$. The chain-rule product is therefore positive, and diminishing returns follow from the concavity of W^* in f . Full proof in EC §EC.1.4. $\square$

The $p=0$ counterfactual in §5.1 confirms this: disabling the workforce multiplier leaves equilibrium CLV exactly invariant to workforce spending. The entire CLV return flows through service capacity, not the operational sensing channel. Standard dashboards, which record technology and workforce returns in separate ledgers, structurally underestimate the financial return to workforce investment.

4.4. Theorem 3: The Burnout Cliff

THEOREM 3 (Burnout Cliff). *Suppose $\sigma > 1$, $p > 0$, and $F_0 < F_{\text{crit}}$ (a surplus equilibrium exists at F_0). Under these conditions the system is bistable in a neighbourhood of F_{crit} . Consider a firm at the surplus equilibrium that implements a step increase in F from F_0 to $F_0 + \Delta F$. Two thresholds govern the outcome:*

(i) *Permanent-collapse onset. For $\Delta F \geq \Delta F_{\text{collapse}}$, no surplus equilibrium exists at the post-upgrade level. The low-trust attractor is the only locally stable state (EC Remark 1); convergence to it is confirmed numerically across the studied initial conditions.*

$$\begin{aligned} \Delta F_{\text{collapse}} &= F_{\text{crit}} - F_0 \\ &= \left(\frac{u}{\delta} [(1-p) + p \frac{f}{f+\mu}] \right)^{1/b} - F_0. \end{aligned}$$

For $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at the post-upgrade level. Whether a given trajectory reaches this equilibrium depends on its initial condition relative to the basin boundary (Figure 2); numerical verification confirms convergence from the pre-shock equilibrium initial condition across the parameter configurations studied (EC §EC.4.2).

(ii) Instantaneous-overload diagnostic. A second, larger threshold marks the boundary above which demand turns positive at $t = 0^+$:

$$\Delta F_{\text{rapid}} = \frac{S_0}{b \cdot F_0^{b-1} \cdot (1 + aT^*)}, \quad (10)$$

where $S_0 = K^*h(W^*) - D(T^*, F_0)$ is the pre-upgrade surplus. Because $\Delta F_{\text{rapid}} > \Delta F_{\text{collapse}}$ (at baseline: 0.40 vs. 0.31), a firm in the zone $\Delta F \in [\Delta F_{\text{collapse}}, \Delta F_{\text{rapid}})$ collapses permanently but through a slow trajectory (G negative at $t = 0^+$, surplus eroded gradually). For $\Delta F \geq \Delta F_{\text{rapid}}$, G turns positive immediately and the system enters a rapid self-reinforcing collapse to the low-trust attractor.

The safe zone is $\Delta F < \Delta F_{\text{collapse}}$; ΔF_{rapid} is a deployment-speed diagnostic, not a safety threshold.

Proof sketch. Part (i) follows directly from Theorem 1: for $F_0 + \Delta F \geq F_{\text{crit}}$, no surplus equilibrium exists; for $\Delta F < \Delta F_{\text{collapse}}$, a new surplus equilibrium exists at $F_0 + \Delta F$, and numerical verification confirms that trajectories originating at the pre-shock surplus equilibrium converge to this new surplus equilibrium across the parameter configurations studied (EC §EC.4.2). For Part (ii), setting the first-order demand increment $\Delta D \approx (1 + aT^*) b F_0^{b-1} \Delta F$ equal to the pre-upgrade surplus S_0 gives ΔF_{rapid} . Under overload the workforce tipping threshold (the W level below which depletion rate exceeds recovery rate) is $W_{\text{tip}} = f/(f + \mu + \sigma G_0^+)$; when $\Delta F < \Delta F_{\text{rapid}}$, $G_0^+ = 0$ at t_0^+ and the system crosses W_{tip} through a slow transient, consistent with the slow-collapse zone. Full proof in EC §EC.1.5. $\square$

Workforce investment f raises F_{crit} and enlarges the safe upgrade zone. Before each deployment phase, the firm should compute $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ as a hard ceiling on planned upgrade size; this ceiling is set entirely by f .

4.5. Proposition 1: Digitalisation-Workforce Complementarity

PROPOSITION 1 (Digitalisation-Workforce Complementarity). *The marginal value of workforce improvement is strictly increasing in the technology base:*

$$\begin{aligned}\frac{\partial K_{\text{eff}}}{\partial W} &= p \cdot K, \\ \frac{\partial^2 K_{\text{eff}}}{\partial W \partial K} &= p > 0.\end{aligned}\tag{11}$$

Since maintaining surplus at higher F requires higher effective capability K_{eff} (that is, either higher K^* through increased u , or higher $h(W^*)$ through increased f), the equilibrium return to workforce investment increases with digitalisation level.

Equivalently, workforce well-being determines the maximum sustainable digitalisation level. At the regime boundary ($S^* \rightarrow 0$), the equilibrium trust approaches zero ($T^* \rightarrow 0$; see Lemma 2, EC §EC.1.2), so:

$$F_{\text{max}}(W) = (K^* \cdot h(W))^{1/b},\tag{12}$$

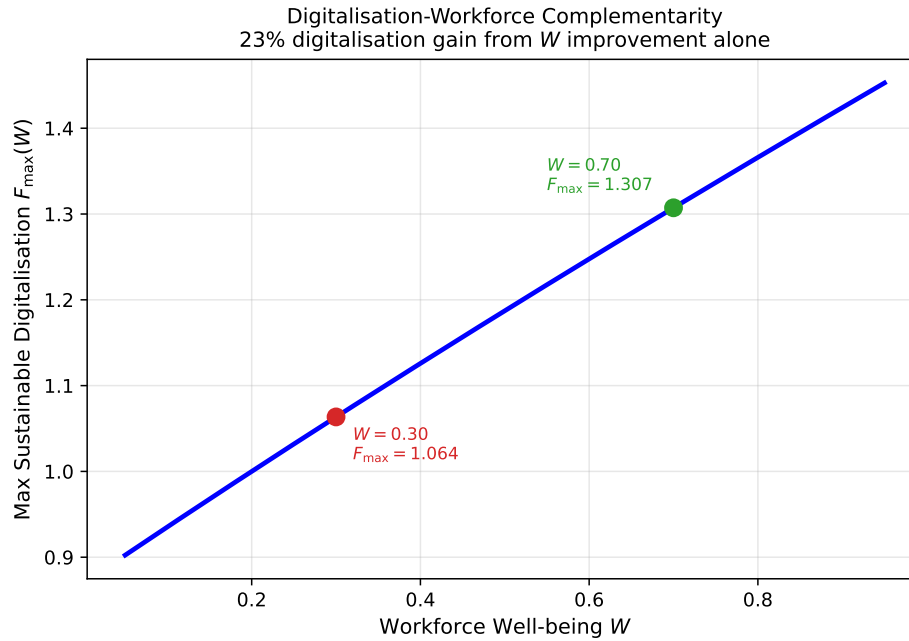
which is strictly increasing in W for $p > 0$. The full implicit differentiation accounting for $T^*(W, F)$ dependence at interior equilibria (where $S^* > 0$) yields a correction term of order $O(aS^*)$ that vanishes as $S^* \rightarrow 0$; the formula (12) is therefore exact at the boundary and an upper bound on F_{max} in the interior: trust-elevated demand means true safe capacity is somewhat lower than this formula implies (EC §EC.1.6).

Proof. The cross-partial follows directly from $K_{\text{eff}} = K[(1-p) + pW]$. The F_{max} formula identifies F_{crit} at $S^* \rightarrow 0$, where $T^* \rightarrow 0$, giving $(K^*h(W))^{1/b}$, strictly increasing in W since $h'(W) = p > 0$; with the full $T^*(W, F)$ correction this is exact at the boundary and an upper bound in the interior (EC §EC.1.6). $\square$

Figure 3 illustrates: the gap between a stressed firm ($W = 0.30$, $F_{\text{max}} \approx 1.063$) and a healthy one ($W = 0.70$, $F_{\text{max}} \approx 1.307$) is 23%, arising entirely from workforce investment.

Under a fixed budget constraint $c_F F + c_f f \leq B$, the CLV-maximising allocation always includes positive workforce investment $f^* > 0$: the all- F corner is strictly dominated whenever the budget reaches or exceeds the Trap threshold (proof in EC §EC.1.7).

Figure 3 Digitalisation-workforce complementarity. The healthy firm ($W = 0.70$) sustains 23% higher $F_{\max}$ than the stressed firm ($W = 0.30$), arising entirely from workforce investment.



5. Computational Study

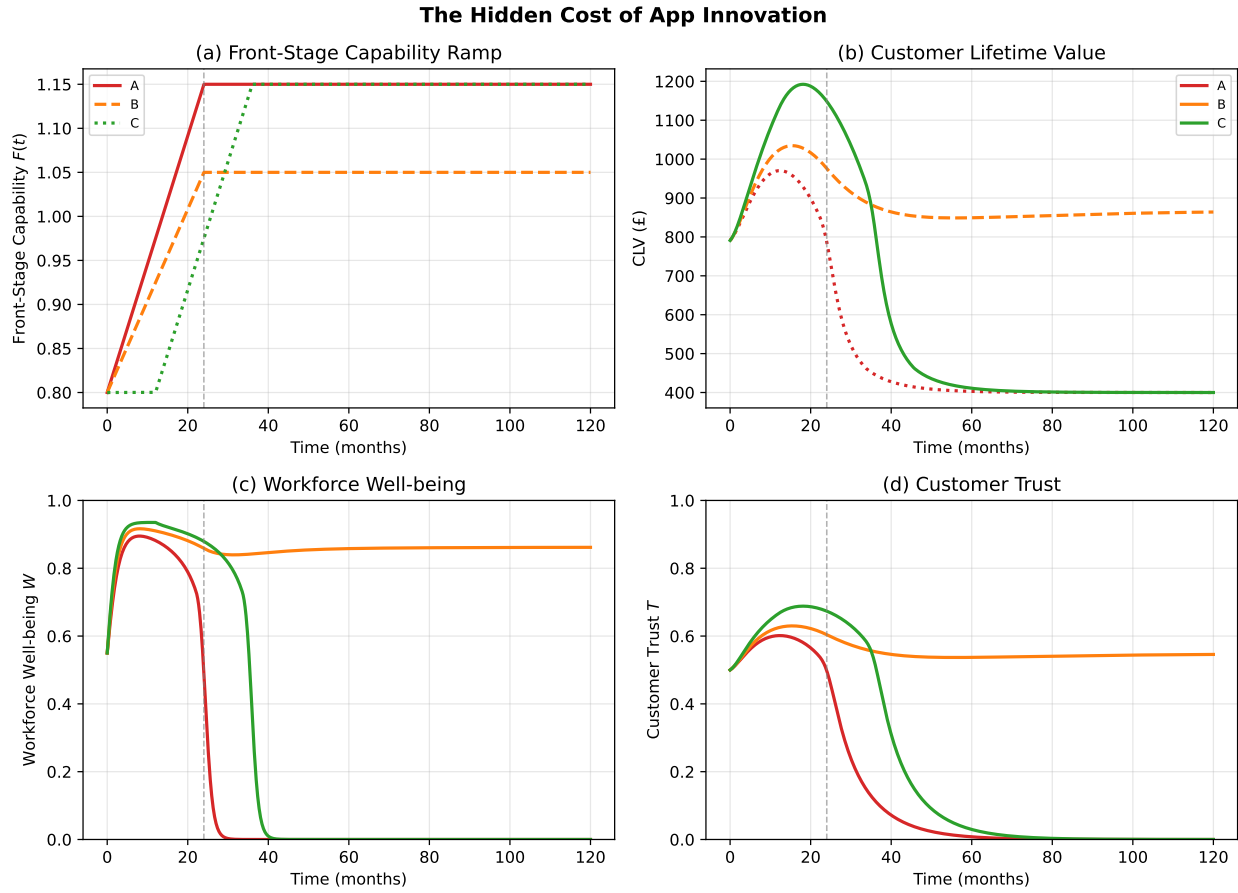
We illustrate the analytical results through calibrated simulations using the baseline parameters in Table 2. CLV follows Gupta et al. (2004): $CLV = w(T)/(1 - \rho(T))$, where $\rho(T)$ is an annual retention probability and $w(T)$ an annualised per-customer margin, with $\rho(T) = 0.80 + 0.18T$ and $w(T) = 80(1 + 0.25 \max(T - 0.15, 0))$. The endpoints $\rho(0) = 0.80$ and $\rho(1) = 0.98$ fall within the 70–90% annual range in Gupta et al. (2004); £80 is an illustrative annual margin for UK omnichannel grocery. All analytical results hold for any CLV increasing in trust (calibration in EC §EC.4.3). All three qualitative phenomena are robust across $b \in [1.1, 2.0]$, $\sigma \in [1.0, 3.0]$, and $p \in [0.25, 0.75]$ (EC Table EC.3).

5.1. The Hidden Cost of App Innovation (Figure 4)

Three deployment strategies span the critical threshold $F_{\text{crit}} \approx 1.11$. All begin from $(K_0, W_0, T_0) = (1.60, 0.55, 0.50)$ with $F_0 = 0.80$. Scenario A ramps aggressively to $F_{\text{end}} = 1.15$ with no workforce investment; Scenario B ramps to $F_{\text{end}} = 1.05$ with concurrent support ($f = 0.06$); Scenario C pre-invests in workforce ($f = 0.10$, months 1–12) before ramping to $F_{\text{end}} = 1.15$.

All three appear profitable in front-stage KPIs (panel a), yet CLV diverges sharply (panel b): by month 24, Scenario C sustains CLV approximately 40% above Scenario A. Workforce collapse

Figure 4 The hidden cost of app innovation. Panel (a): front-stage KPIs rise for all three scenarios. Panel (b): CLV diverges sharply by month 24. Panels (c,d): workforce collapse precedes trust collapse in the no-investment scenario.



precedes trust collapse (panels c, d), confirming the cross-domain cost invisible in the technology business case.

5.2. Basin-of-Attraction Structure

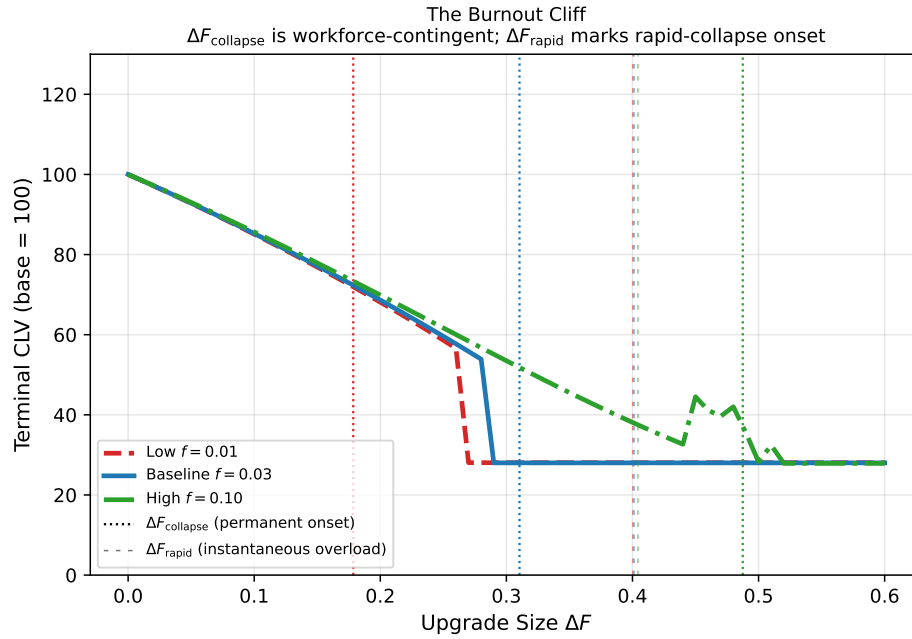
Figure 2 maps the basin boundary across the joint space of W_0 and F . The boundary shifts rightward as W_0 increases: a firm near the basin boundary can increase its safety margin by raising workforce investment f , which shifts F_{crit} rightward without touching the technology stock.

5.3. The Burnout Cliff (Figure 5)

We sweep $\Delta F \in [0, 0.60]$ for three workforce-investment scenarios: neglected ($f = 0.01$), baseline ($f = 0.03$), and invested ($f = 0.10$), each starting from its surplus equilibrium at $F_0 = 0.80$. Varying f shifts $F_{\text{crit}}(f)$ from 0.98 to 1.29, widening $\Delta F_{\text{collapse}}$ from 0.18 to 0.49.

Figure 5 shows the three normalised CLV curves. For the neglected firm ($f = 0.01$), the safe zone collapses to $\Delta F < 0.18$; for the invested firm ($f = 0.10$), it extends to $\Delta F = 0.49$. The slow-

Figure 5 The Burnout Cliff. Normalised terminal CLV versus upgrade size ΔF for three workforce-investment levels. Dotted verticals: $\Delta F_{\text{collapse}}$ (0.18, 0.31, 0.49). The neglected firm ($f = 0.01$) exhausts its safe zone at $\Delta F = 0.18$; the invested firm ($f = 0.10$) retains headroom to $\Delta F = 0.49$.



collapse zone between $\Delta F_{\text{collapse}}$ and ΔF_{rapid} is the diagnostic blind spot: dashboards report continued growth while the trust trajectory has already turned negative. For the well-investing firm, this zone disappears entirely (EC §EC.1.5).

6. Empirical Evidence

The model yields three testable predictions at the firm–year level: (i) a positive technology–workforce interaction ($K \times W > 0$); (ii) an inverted-U between e-commerce intensity and customer satisfaction; (iii) a positive effect of workforce improvement on satisfaction that increases with technology intensity. We test each using panel data from two independent geographies.

6.1. Data and Sample

US sample. We construct a balanced panel of 13 publicly listed US retailers observed annually from 2010 to 2024. Customer satisfaction (T proxy) is measured by the American Customer Satisfaction Index (ACSI; Mittal et al. 1998), a nationally representative survey scored 0–100 (source: theacsi.org). Workforce well-being (W proxy) uses Glassdoor overall ratings (1–5 scale). Technology investment (K proxy) is capital expenditure intensity (CapEx/Revenue) extracted from SEC EDGAR 10-K filings. E-commerce share of retail sales is from the Federal Reserve Economic Data series ECOMPCTSA. Revenue (size control), SGA intensity, and net margin are from

EDGAR. After listwise deletion of missing observations, the regression sample contains 131 complete firm–year observations across 13 firms and 12 years.

EU sample. An independent replication panel covers 11 European retailers across six countries (UK, France, Germany, Spain, Sweden, Netherlands) from 2019 to 2023. Customer satisfaction uses Trustpilot scores (1–5 scale; rescaled to 0–100 for comparability). Glassdoor ratings proxy workforce well-being. Financial data are from Yahoo Finance. Because financials are reported in mixed currencies (GBP, EUR, SEK), we use within-year asset percentile rank as the size control in cross-sectional specifications and log revenue only in within-firm (fixed-effects) specifications where currency is absorbed. E-commerce share uses UK Office for National Statistics internet sales data for UK firms and Eurostat data for continental retailers. The complete panel contains 20 firm–year observations with all three proxies present.

6.2. Identification Strategy

All variables are standardised to mean zero and unit standard deviation before regression. We estimate three specifications mapping to the three model predictions:

Test 1 (Complementarity $\rightarrow$ Proposition 1).

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$$T_{it} = \alpha_i + \gamma_t + \beta_1 K_{it}^z + \beta_2 W_{it}^z + \beta_3 (K \times W)_{it}^z + \mathbf{X}_{it} \boldsymbol{\delta} + \varepsilon_{it}. \quad (13)$$

Prediction: $\beta_3 > 0$. Two-way fixed effects (firm α_i and year γ_t) absorb time-invariant firm heterogeneity and common shocks. Controls $\mathbf{X}_{it}$ include log revenue, and in extended specifications, SGA intensity and net margin.

Test 2 (Front-Stage Trap $\rightarrow$ Theorem 1).

$$T_{it} = \alpha_i + \beta_1 \text{Ecom}_{it} + \beta_2 \text{Ecom}_{it}^2 + \mathbf{X}_{it} \boldsymbol{\delta} + \varepsilon_{it}. \quad (14)$$

Prediction: $\beta_1 > 0$, $\beta_2 < 0$ (inverted-U). Turning point: $\text{Ecom}^* = -\beta_1/(2\beta_2)$.

Test 3 (Good Jobs Buffer $\rightarrow$ Theorem 2).

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$$\Delta T_{it} = \alpha_i + \gamma_t + \beta_1 \Delta W_{it} + \beta_2 (\Delta W \times K)_{it} + \mathbf{X}_{it} \boldsymbol{\delta} + \varepsilon_{it}. \quad (15)$$

Prediction: $\beta_2 > 0$ (workforce improvement matters more at high technology intensity).

Inference corrections. With only $N_c = 13$ (US) and $N_c = 11$ (EU) firm clusters, standard clustered standard errors are biased downward. Following Cameron et al. (2008), we report p -values from the $t(N_c - 1)$ distribution throughout and supplement with cluster-robust (CR2) finite-sample corrections (Bell and McCaffrey 2002) and Webb six-point wild cluster bootstrap (Webb 2023) in the EC. All corrections are applied uniformly across both geographies.

6.3. Results

Table 3 reports the US results.

Table 3 US regression results. Dependent variable: ACSI score (0–100). All p -values use $t(N_c - 1)$ correction with $N_c = 13$ –17 clusters. Significance: * $p < 0.01$, ** $p < 0.05$, * $p < 0.10$.**

	1a: OLS	1b: Firm FE	1c: Two-way FE	1d: FE + controls
<i>Test 1: Complementarity</i>				
K^z	0.2903	−0.2153	−0.4218	−0.3102
W^z	1.2506	0.1087	0.3181	0.4553
$K \times W^z$	1.9533	−0.6342**	−0.5615*	−0.7649
N	131	131	131	83
<i>Test 2: Front-Stage Trap</i>				
	2a: Linear	2b: Quadratic	2c: Quad+W	
Ecom	0.4379	0.4400	0.0284	
Ecom ²		−0.0279**	−0.0055	
Turning point		7.9%	2.6%	
N	216	216	190	
<i>Test 3: Good Jobs Buffer</i>				
	3a: ΔW only	3b: $\Delta W \times K$	3c: + controls	
$\Delta W \times K$		−41.40	−132.37	
N	173	118	84	

N varies across tests because each requires different variable coverage: Test 1 requires ACSI, Glassdoor, and CapEx; Test 2 requires only ACSI and e-commerce share; Test 3 uses first-differenced variables, losing one observation per firm.

Test 1 (Complementarity). The cross-sectional (OLS) estimate of β_3 is positive (+1.95), consistent with the model's prediction that firms with jointly higher technology and workforce investment enjoy higher satisfaction. However, the within-firm estimates (Models 1b–1d) are negative and sometimes significant, suggesting that year-to-year co-movements in K and W within the same firm do not raise satisfaction. This reversal is consistent with low within-firm variation in Glassdoor ratings (within-firm SD is only 30% of total SD): firm fixed effects absorb the cross-sectional signal that drives complementarity.

Test 2 (Front-Stage Trap). The quadratic specification (2b) yields $\hat{\beta}_2 = -0.028$ ($p = 0.028$ under $t(16)$), confirming the predicted inverted-U. The estimated turning point of 7.9% e-commerce share corresponds to the early-to-mid 2010s transition, broadly consistent with the model's $F_{\text{crit}} \approx 1.11$ threshold.

Test 3 (Good Jobs Buffer). The interaction $\Delta W \times K$ is negative in both specifications, contrary to the prediction. The first-differenced design, combined with noisy annual Glassdoor ratings and few complete observations ($N = 84\text{--}118$), leaves this test underpowered.

EU replication. The European sample provides an independent out-of-sample test. Using Trustpilot (1–5 scale) as the outcome, the between-effects estimate of β_3 is positive (+0.47) and significant under pooled cluster-robust inference (CR2: $p = 0.045$ at $t(10)$). However, the conservative HC3 between-effects estimate ($p = 0.53$) and Webb bootstrap ($p = 0.62$) do not reject zero, reflecting the extreme power limitation of $N_c = 11$ clusters. The qualitative sign consistency across geographies—positive $K \times W$ in cross-section—provides suggestive, though not conclusive, support for complementarity.

6.4. Limitations of the Empirical Analysis

Four limitations warrant emphasis. First, with $N_c \in \{11, 13\}$ clusters, clustered inference is at the boundary of asymptotic validity; we report wild bootstrap p -values in the EC for all key coefficients. Second, Glassdoor ratings exhibit low within-firm temporal variation, making the complementarity interaction difficult to identify within firm over time; between-effects estimates (which exploit cross-sectional variation) are more informative but face omitted-variable concerns. Third, the EU outcome (Trustpilot) is complaint-biased and operates on a different scale than ACSI; while we rescale to 0–100 for coefficient comparability, measurement equivalence is not guaranteed. Fourth, the data are at the firm–year level; the model's predictions concern service-capacity dynamics that are most cleanly testable with store-level panel data and exogenous technology rollout events. These

results are best interpreted as suggestive evidence consistent with the model's qualitative predictions, pending the store-level identification strategy outlined in §7.2.

7. Discussion

7.1. Contributions and Managerial Implications

The three theorems together provide the formal micro-foundation that connects Ton's (2014) four operational choices to each other through the dynamics of K_{eff} . Her prescription to “operate with slack” corresponds to the positive workforce investment rate f : operational slack keeps W from depleting under moderate overload, and Theorem 2 quantifies the CLV return to doing so. “Standardise and empower” corresponds to a lower overload sensitivity σ : well-designed processes reduce the per-unit depletion rate and enlarge the safe upgrade zone of Theorem 3. “Offer less” maps directly onto keeping F below F_{crit} : the model gives this demand-side lever a computable threshold rather than a qualitative prescription. “Cross-train” maps onto a higher complementarity parameter p : cross-trained workers raise effective capability $h(W)$ at any given engagement level, increasing the marginal return to both technology and workforce investment simultaneously (Proposition 1). The model extends Oliva–Sterman with customer trust, a front-/back-stage domain split, and digital scaling: three extensions absent from the quality-erosion framework, now formalised in a single model that, to the best of our knowledge, provides the first dynamic formalisation of the service profit chain (Heskett et al. 1994) and Good Jobs Strategy (Ton 2014) with computable boundary conditions.

Managerial implications. Three implications follow. First, the maximum safe rollout speed is a workforce constraint (Theorem 3): the firm should verify that $\Delta F_{\text{collapse}} = F_{\text{crit}}(f) - F_0$ exceeds the planned upgrade size. Second, workforce well-being is a leading indicator: W reaches crisis levels 6–21 months before trust does (EC Table EC.4). Third, the more digital the firm becomes, the more it should invest in its people (Proposition 1): conventional ROI calculations underestimate returns because the dominant capacity-multiplier channel is invisible in standard dashboards (Theorem 2).

7.2. Limitations and Directions for Future Research

Model limitations. The model treats F as exogenous; endogenising it would enable optimal deployment sequencing. The omission of a direct $W \rightarrow D$ channel makes our results a conservative lower bound on true complementarity.

Empirical limitations. The firm-level panel analysis (§6) provides suggestive but not definitive support. With only 13 (US) and 11 (EU) firm clusters, all clustered inference operates at the boundary of asymptotic validity. The complementarity interaction reverses sign under firm fixed effects,

consistent with limited within-firm temporal variation in Glassdoor ratings rather than absence of the effect. Store-level panel data with exogenous technology rollout events—the gold-standard identification strategy—would permit sharper separation of the capacity-multiplier channel from confounds.

Extensions. Three, in order of expected impact: (i) store-level identification of p using technology rollout timing as instruments; (ii) a competitive multi-firm extension where workforce poaching creates strategic complementarity; (iii) stochastic demand forcing, which would transform the Burnout Cliff into a probabilistic threshold.

8. Conclusion

Standard operational dashboards measure the front-stage domain that invests in technology, not the back-stage workforce that fulfils the resulting demand and absorbs the consequences. This disconnect is not merely a measurement problem; it is a structural feature of omnichannel operations that makes cross-domain value destruction structurally predictable from two observable managerial choices: the workforce investment rate f and the digitalisation level F . This paper provides the formal instrument for both.

Three results follow that, to our knowledge, no existing single formulation produces. The Front-Stage Trap (Theorem 1): a computable digitalisation threshold $F_{\text{crit}} \approx 1.11$ beyond which additional front-stage investment destroys CLV by overloading back-stage workers, a sign reversal impossible in single-domain models. The Good Jobs Buffer (Theorem 2): workforce investment raises CLV through a capacity-level channel confirmed exact by the $p=0$ counterfactual, which leaves CLV invariant to workforce spending and isolates the entire gain to service capacity. The Burnout Cliff (Theorem 3): the maximum safe upgrade size ranges from $\Delta F = 0.18$ for a workforce-neglecting firm to $\Delta F = 0.49$ for an investing firm, an approximately 172% difference; the neglected firm retains approximately 37% of the safe deployment headroom.

These results give operations teams two computable numbers standard dashboards do not report: a digitalisation ceiling (F_{crit}) and a maximum safe rollout size ($\Delta F_{\text{collapse}}$). Firm-level panel evidence from 13 US and 11 EU retailers provides suggestive support: the technology–workforce interaction is positive in cross-section (both geographies), and an inverted-U between e-commerce share and customer satisfaction aligns with the predicted digitalisation ceiling. Store-level identification of p using exogenous technology rollout events remains the most important empirical follow-up. The broader implication extends beyond omnichannel retail: wherever automated front-stage processes generate demand on human back-stage operations, workforce investment is a structural requirement, not a social concession.

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